

QKD CODING EXAMPLE

Actual quantum programming for QKD like BB84 would require interaction with quantum hardware or simulators and is complex. However, we can conceptualise a very simplified pseudo-code snippet that outlines the core logic of the BB84 protocol:

Simplified Pseudo-code for BB84 Protocol Concept

```
# Alice's preparation
alice_bases = random_sequence_of_bases()
alice_states = random_sequence_of_qubit_states()
```

```
# Bob's measurement
bob_bases = random_sequence_of_bases()
bob_measurements = measure_qubits(alice_states, bob_bases)

# Compare bases and keep matching results
shared_key = []
for i in range(len(alice_bases)):
    if alice_bases[i] == bob_bases[i]:
        shared_key.append(bob_measurements[i])
```

Imagine Alice wants to send a secure message to Bob without Eve intercepting it. Using QKD, Alice sends a series of qubits to Bob, each encoded in a random quantum state. Bob attempts to measure these qubits using random bases. After the transmission, Alice and Bob publicly compare their choice of bases (but not the results of their measurements) and keep only the results obtained from matching bases to form a secret key. Any attempt by Eve to eavesdrop would involve measuring the qubits, thus altering their states and revealing her presence.

As quantum computing continues to evolve, its implications for cryptography and security become increasingly significant. Quantum Key Distribution (QKD) and Post-Quantum Cryptography (PQC) represent two critical fronts in the ongoing effort to safeguard information in the quantum age.

Quantum Key Distribution (QKD)

Quantum Key Distribution leverages the principles of quantum mechanics to achieve secure communication, ensuring that any attempt at eavesdropping can be detected by the legitimate parties involved.

How QKD Works

QKD utilises quantum mechanics to securely distribute keys between two parties, traditionally named Alice and Bob. The process involves several steps:

- **Key Transmission:** Alice sends Bob a series of qubits, each encoded in one of several possible quantum states. This transmission utilises quantum channels that could be optical fibers or even free-space links.
- **Basis Selection and Measurement:** Both Alice and Bob randomly choose measurement bases (e.g., rectilinear or diagonal for the BB84 protocol) to measure the qubits. They then publicly share their choice of bases without revealing the measurement outcomes.

- **Sifting:** Alice and Bob discard any bits where they used different bases, keeping only the bits derived from qubits measured in the same basis.
- **Error Rate Estimation:** They compare a subset of their bits to estimate the error rate in their transmission, which helps detect potential eavesdropping.
- **Privacy Amplification and Error Correction:** Finally, they use classical post-processing techniques to correct errors and enhance the security of their key.

Implementations and Challenges

Real-world implementations of QKD have successfully demonstrated its feasibility over both fibre optic networks and through atmospheric transmission. However, challenges remain in terms of range, transmission rates, and scalability. Efforts to overcome these limitations include the development of quantum repeaters and the utilisation of satellite-based QKD to achieve global secure communication networks.

Post-Quantum Cryptography (PQC)

